

SHIPRAG — System Architecture & Workflows

Complete Technical Architecture & Microservice Component Walkthrough

1. System Architecture Overview

SHIPRAG is an enterprise-grade Retrieval-Augmented Generation (RAG) platform designed for multi-tenant document ingestion, token window chunking, dense vector storage with pgvector, and hybrid search generation.

Layer / Stage	Components & Responsibilities	Target Port / Endpoint
Client / Ingestion Layer	CLI Tool, Web Dashboard, GitHub Push Webhooks	Client Request
Auth Service	JWT Bearer Validation, sk-shiprag-... API Keys, Multi-Tenant Isolation	:8001 (/api/v1/auth)
Ingestion Service	Multi-Format Document Parsing (.md, .pdf, .docx, .html), Redis Stream Producer	:8002 (/api/v1/ingest)
Embedding Worker	Redis Stream Consumer, Tiktoken Sliding Window Chunking, Vector Embeddings (1536 dims)	:8003 (Worker Daemon)
Persistence Layer	PostgreSQL + pgvector (Dense Vectors), Redis Cache (Tokens & Metadata)	:5432 / :6379
Retrieval & RAG Service	Hybrid Search (70% Dense Vector + 30% BM25 Keyword), LLM Generation (claude-sonnet-4-6), SSE Stream	:8004 (/api/v1/retrieval)

2. `demo\_backend.py` Execution Workflow Sequence

Step 1: Configuration Loading	Loads central configuration settings from app.core.config (Embedding model: <i>text-embedding-3-small</i> , Chunk size: <i>500 tokens</i> , Overlap: <i>50 tokens</i> , Top K: <i>5</i>).
Step 2: Token Chunking	Processes raw input document through app.core.chunking. Applies sliding window tokenizer to slice content cleanly.
Step 3: Dense Embedding	Passes tokenized text chunks into app.core.embeddings to produce 1536-dimensional float vector representations.
Step 4: Vector Storage & Retrieval	Interacts with app.core.vectorstore to perform hybrid cosine-similarity search against stored pgvector embeddings.
Step 5: Service Routing Audit	Verifies health status and endpoint reachability for microservices across ports :8001 through :8004.

3. Microservices Architecture Matrix

Service	Port	Key Responsibilities	Tech Stack
Auth Service	:8001	Tenant Auth, JWT tokens, sk-shiprag API Key validation	FastAPI, PyJWT, SQLAlchemy
Ingestion Service	:8002	Doc parsing, GitHub push webhooks, Redis producer	FastAPI, PyPDF2, Redis
Embedding Worker	:8003	Redis consumer, Tiktoken chunker, Vector upserts	Python, Tiktoken, pgvector
Retrieval Service	:8004	70/30 Hybrid Search, LLM generation, SSE streaming	FastAPI, Anthropic API, Redis